Advaith **Sriram**

ROBOTICS ENGINEER

About me

Robotics Master's student at **EPFL, Switzerland**, with expertise in robotic control, perception, AI-driven motion planning and reinforcement learning (RL). Co-first author of an ICRA 2024 publication and lead developer of an ABB GoFa system exhibited at the Swiss Design Awards 2025. Having lived and worked in both Switzerland and Japan, I have developed a strong understanding of diverse engineering cultures and thrive in international, interdisciplinary teams. Seeking a full-time robotics engineering role in autonomy, vision, or manipulation.

Portfolio

advaithsriram.github.io/

advaith-sriram

advaith-sriram

Interests

- Active Perception and Manipulation
- Legged Robotics
- Reinforcement Learning

Contact Details

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Lausanne, Switzerland

Education

Sep 2024 - Sep 2026

M.Sc. in Robotics

École Polytechnique Fédérale de Lausanne (EPFL) — **Lausanne, Switzerland**

- Relevant Coursework: Legged Robots, Computational Motor Control, Aerial Robotics, Robotics for Manipulation, Model Predictive Control, Convex Optimization.
- Minor in Data Science: Machine Learning, Applied Data Analysis, Computer Vision, Image Analysis.
- Teaching Assistant: *Legged Robots*, *Mobile Robotics*

Sep 2019 - Sep 2023

B.E. in Mechanical Engineering

Waseda University — **Tokyo, Japan**

- Specialization in **Robotics**: Human-Robot Co-Existence, *Sugano Laboratory*.
- Researched Large-Object Human-to-Robot Handover as part of Japan Science and Technology Agency's (JST) Moonshot Program Goal 3.
- Minor in Computer Science and Communications Engineering

Professional Experience

Oct 2023 - Sep 2024

Mechatronics Engineer

Daimler Trucks Asia — **Kanagawa, Japan**

- Integrated a Position and Orientation System (POS-LV) sensor to extract data for accurate road geometry using ROS 2.
- Converted ROS 1 libraries to ROS 2, reducing future maintenance load and improving cross-platform compatibility.
- Contributed to vehicle control R&D for a Highway Level-4 autonomous trucking program.

Internship Experience

Mar – Sep 2026

Robotics Engineering Intern (M.Sc. Thesis)

Cybernetics Laboratory (CynLr) SA — **Lausanne, Switzerland**

- Developing an active perception and view planning framework for robotic manipulation.
- Collaborating with industry and EPFL Learning Algorithms and Systems Laboratory (LASA) for joint supervision and research development.

May – Sep 2023

Research Intern

Beckhoff Automation K.K. — **Tokyo, Japan**

- Implemented a cryptographic security layer on an EtherCAT industrial controller using TwinCAT, gaining hands-on experience with real-time control systems and industrial automation hardware.
- Collaborated with the University of Electro-Communications to research ElGamal encryption, leveraging their expertise to enhance data security on robotic hardware.

Publications

May 2024

Overcoming Hand and Arm Occlusion in Human-to-Robot Handovers: Predicting Safe Poses with a Multimodal DNN Regression Model

2024 IEEE International Conference on Robotics and Automation (ICRA)

- Primary researcher and co-first author. Conducted all experiments, data collection, and result analysis. Research originated from my Bachelor's thesis at Waseda University's Sugano Laboratory.
- **DOI**: 10.1109/ICRA57147.2024.10610777

Languages

- **English** (Native)
- **Tamil** (Native)
- **Japanese** (**JLPT N3** Level)
- **French** (A2 - Level)

Technical Skills

Languages: Python, C/C++, C#, MATLAB

Robotics: ROS 2, Path Planning, Reinforcement Learning, Model Predictive Control

AI/ML: PyTorch, TensorFlow, VAEs

Vision: OpenCV, DNNs

Simulation: PyBullet, Unity, Gazebo, Webots

Platforms: Git, Docker, Linux, Jupyter

Hardware: ABB GoFa, Crazyflie

Academic Projects

Sep 2025 - Jan 2026

Next-Best-View Aerial Swarms for Multi-Viewpoint Monitoring

Semester Project @ Laboratory of Intelligent Systems (LIS), EPFL

- Designed and evaluated a multi-drone monitoring framework comparing swarm-based control (Olfati–Saber) with Next-Best-View (NBV) planning for 3D scene observability.
- Implemented a full simulation pipeline in Unity, integrating depth sensing, point cloud fusion, and NBV planning guided by a PoinTr-based shape completion model.
- Proposed a mesh visibility metric to quantitatively assess surface observability without full reconstruction, and demonstrated trade-offs between coverage efficiency, flight time, and data redundancy.
- **Tech Stack**: Python, Unity, C#, OpenCV

Feb - Jun 2025

AI and Music-Driven Choreography for ABB GoFa Robot

Semester Project @ REHAssist Lab, EPFL (Collaboration with ABB)

- Created an interactive installation with the ABB GoFa CRB 15000 robot arm, mapping instrumental music to expressive robot motion using a custom-trained Variational Autoencoder (VAE)
- Deployed the system using ROS 2 Humble for joint control and integrated a hand-following element using MediaPipe for real-time human interaction.
- Exhibited the final robotic system at the Swiss Design Awards 2025, Basel.
- **Tech Stack**: Python, C++, ROS 2, VAE, ABB RobotStudio Suite.

Sep 2024 - Jan 2025

Reinforcement Learning for Legged Locomotion Simulation

Course Project (Legged Robots)

- Designed action, observation, and reward spaces from scratch for a simulated quadruped (Unitree A1) to achieve stable locomotion over slopes, stairs, and gaps without a reference gait.
- Applied domain randomization via PPO (Stable-Baselines3) to improve real-world transferability, demonstrating robust performance across varied terrain conditions.
- **Tech Stack**: Python, PyBullet, Stable-Baselines3 (PPO), PyTorch, OpenAI Gym.

Extracurriculars

Sep 2021 – Aug 2023

President, Co-Founder

Google Developer Groups on Campus - Waseda University — **Tokyo, Japan**

- **Co-founded and led** a developer group of 75 student members, achieving significant event participation that engaged over 150 students.
- Organized an international hackathon spanning 8 universities across Japan and South Korea with 100+ participants.
- Enhanced the club's campus presence and *facilitated valuable internship opportunities* for members.